

Homework 5

Due Thursday, March 27

1. (6 pts.) Problem 6.B from the textbook. Please assume that the operators in question are normalized by the condition

$$\mathrm{Tr}(E_\alpha E_{-\alpha}) = \lambda > 0,$$

where the trace is in the adjoint, and λ is the same for all the root vectors α .

2. (14 pts.) Problem 7.B from the textbook. Note that the problem has two parts: decomposition of the 3-dimensional representation and of the adjoint. Note also that in this problem T_a denote the matrices of the defining, rather than adjoint, representation of $\mathrm{SU}(3)$, see Eq. (7.6) of the textbook.

A good way to approach this problem is through the highest weight (HW) construction. We know how the generator T_3 of the $\mathrm{SU}(2)$ subalgebra acts in both of the $\mathrm{SU}(3)$ irreps in question. So, we can find in each case the eigenstate(s) of T_3 with the largest eigenvalue and, starting from those as HW states, the entire corresponding irreps of the $\mathrm{SU}(2)$. Then, we can similarly consider the action of T_3 on the remaining states.